

Titouan Renard,

Political Economy Student

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Political economy student with a background in robotics and artificial intelligence, I have strong foundations in mathematics, strong abstract reasoning skills, and extensive programming experience. Before political economy, I was mostly interested in optimal control, reinforcement learning and learning theory. Now I am mostly interested in international economics, crises and complexity economics.

Education

Master in Political Economy of Capitalism , Geneva, Switzerland UNIGE	2024-2026
Master in Economics , Barcelona, Spain (Exchange Semester) Universitat de Barcelona	2025
Master in Robotics with a Minor in Data Science , Lausanne, Switzerland EPFL, STI and IC faculties	2023
Bachelor in Microengineering , Lausanne, Switzerland EPFL, STI faculty	2020
Bilingual Maturité , Neuchâtel, Switzerland French-English bilingual high-school, Lycée Denis de Rougemont, Major in Physics and Applied Mathematics	2016

Past Projects

Measuring Global Value Chains Integration Using Embodied Labour , seminar paper for the course "Global Sustainability" at Universitat de Barcelona, under the supervision of Prof. Mònica Serrano. Project Graded 10/10 Investigated the labor contribution relative to final demand of different sectors/countries using inter-country input-output (ICIO) tables. Project report online.	2025
Swiss Financial Diplomacy in the Interwar Period, Preserving Bondholders' Interests in the Context of the Renegotiation of the Ottoman Public Debt (1919-1928) , seminar project at UNIGE under the supervision of Prof. Juan Flores Zendejas. Project Graded 6/6 Investigated the collaboration between Swiss Diplomats and Bondholders using diplomatic archival sources. Project report online.	2025
Provable Convergence Guarantees for Constrained Inverse Reinforcement Learning , master project at SYCAMORE (Systems Control and Multiagent Optimization Research) at EPFL, under the supervision of Prof. Maryam Kamgarmour. Project Graded 6/6 Proved convergence guarantees for a constrained IRL algorithm, led to a publication in IEEE CDC.	2023
Learning Motor Policies with Time Continuous Neural Networks , semester project supervised by two labs, BIOROB (Biorobotics) and LCN (Laboratory of Computational Neuroscience) at EPFL, under the supervision of Prof. Auke Ijspeert and Prof. Wulfram Gerstner. Project Graded 6/6 Work on the use of time continuous neural networks (Liquid Time-Constant Neural Networks) to solve control problems using reinforcement learning methods. Project report online.	2022
Power-Optimal Trajectory Generation for Airborne Wind Energy , semester project at LA3 (EPFL), under the supervision of Prof. Colin Jones. Project graded 6/6 . Worked on a strongly non-linear and non-convex optimization problem, implemented an appropriate iterative solver method to deal with the non-convexity in Matlab. Project report online.	2021

Experience

Internships

Research Intern at SYCAMORE (Systems Control and Multiagent Optimization Research) at EPFL, under the supervision of Maryam Kamgarmour Worked on the sample complexity of Inverse Reinforcement Learning (published work).	2023
Course Project Design at the Laboratory of Computational Neuroscience, under the supervision of Alireza Modirshanechi Designed the course project for the Artificial Neural Networks and Reinforcement Learning course at EPFL.	2023
Machine-Learning Intern at 3HLE Robotics and Automation	2022-2023

Worked on deep-learning models for <i>6D pose estimation</i> from rgb-d sensor data in the context of industrial robotics manipulation, under the supervision of Laurent Winkler	
Summer Internship , at EPFL's Birorobotics Lab, under the supervision of Alessandro Crespi.	2018
Implementation of a simple vision-based localization system using C++ and OpenCV.	
Student-Assistant at EPFL	
Artificial Neural Networks and Reinforcement Learning	2022-2023
For the Computer Science and Data Science students.	
"Physique Générale 1" (Newtonian Mechanics)	2018-2021
For the PH-101(b) (Microengineering, Electrical Engineering and Material Sciences sections) and PH-101(d) courses (for the Mechanical Engineering section).	
"Microinformatique" (basics of embedded systems programming)	2019
For the Microengineering students.	
Basics of Robotics	2018
For the Microengineering students.	
Material Sciences Practicals	2018
For the Microengineering and Material sciences students.	
Other	
Webmaster for Institut de Psychanalyse Charles Baudouin	2025-present
Development, maintenance and updating of the institute's website.	

Skills

Programming, in C, C++, Javascript, Matlab, Python and R.

Math and Applied Math, Probabilities, Linear Algebra, Calculus, Econometrics. Strong abstraction capacity, ability to mathematically formulate a concrete problem.

Soft Skills, public speaking, writing and team management learned in the context student associations, graphic design, French (native proficiency), English (full proficiency), German (B2), Spanish (B1)

Volunteer Experience

Unipoly , student's association for sustainability at EPFL.	2019-2022
Board member, then president of the association.	
European Youth Parliament , Switzerland	2015-2016
English language public speaking experience.	

Publications

- [1] «L'allègement budgétaire» pour qui? Les contradictions du programme d'austérité suisse
Marc Morgan, Romain Desgraz, Adnan Chatila, Titouan Alexis Arthur Renard, Miléna Locher,
Political Economy Working Papers 7/2025 (Dec. 11, 2025), p. 35
URL: <https://archive-ouverte.unige.ch/unige:189618>
- [2] *Provable Convergence Guarantees for Constrained Inverse Reinforcement Learning*
Titouan Renard
2023
Available at <https://infoscience.epfl.ch/record/304492?ln=en>
- [3] Convergence of a model-free entropy-regularized inverse reinforcement learning algorithm
Titouan Renard, Andreas Schleginhäufen, Tingting Ni, Maryam Kamgarpour
2024 IEEE 63rd Conference on Decision and Control (CDC), 2024
DOI: [10.1109/CDC56724.2024.10886001](https://doi.org/10.1109/CDC56724.2024.10886001)
- [4] L'urgence climatique est encore et toujours à l'ordre du jour
Karine Clerc, Titouan Renard, Sébastien Maillard
Voix Populaire (2023)
URL: <https://voixpopulaire.ch/2023/02/07/lurgence-climatique-est-encore-et-toujours-a-lordre-du-jour/>
- [5] Dominique Bourg et al.
Désobéir pour la Terre, Une défense de l'état de nécessité,
Presses Universitaires de France, 2021,
ISBN: 9782130830368.
- [6] Raphaël Arlettaz et al.

Tumulte postcorona, Les crises, en sortir et bifurquer,
Éditions d'en bas, 2020,
ISBN: 9782829006173.